MosaicMem: Hybrid Spatial Memory for Controllable Video World Models

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Research Questions

- How can spatial memory mechanisms for video world models combine the geometric consistency of explicit 3D memory with the dynamic flexibility of implicit memory?
- Can patch-level spatial memory with 3D lifting and implicit conditioning achieve better camera control and dynamic object modeling than existing paradigms?
- How can warped positional encoding and latent alignment improve memory retrieval accuracy in video diffusion models?

Core Idea

Existing spatial memory approaches for video world models face a fundamental tradeoff: explicit 3D memory (point clouds, 3D Gaussians) provides geometric consistency but struggles with dynamic objects, while implicit memory (posed frames) handles dynamics but suffers from camera drift and redundancy. MosaicMem bridges this gap by using patches as memory units - lifting them into 3D for reliable localization (like explicit methods) but conditioning generation through attention mechanisms (like implicit methods). The key insight is that patches offer the right granularity: more structured than implicit frames for geometric alignment, yet more flexible than explicit point clouds for dynamic scenes.

Key Formula

Warped RoPE Reprojection

$$(u', v') = \Pi\left(\boldsymbol{K}_j \boldsymbol{T}_j \boldsymbol{T}_i^{-1} \boldsymbol{K}_i^{-1}(u, v, D)\right)$$

Projects patch coordinates (u,v) with depth D from source camera (Ki, Ti) to target camera (Kj, Tj). Π denotes perspective projection. This enables geometric correspondence for aligning memory patches across viewpoints, addressing the core challenge of patch-level spatial consistency.

Key Figure

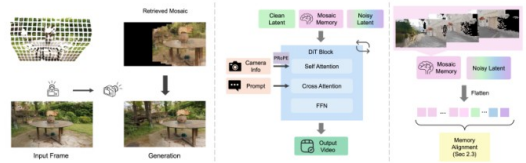

Fig. 2: Method overview. Left: MosaicMem lifts patches into 3D, then gathers and stitches them in the target view like a mosaic. Middle: Architecture overview. Camera motion is controlled jointly by MosaicMem retrieval and PRoPE conditioning. Right: Retrieved mosaic patches are flattened and concatenated to the token sequence as conditioning, while alignment errors were solved by warping.

camera trajectory, faithfully retrieves spatial memory from real observations or previously generated clips, and renders scene dynamics as well as unseen content consistent with the text prompts.

Overview. Our method builds on text+image-to-video (TI2V) models by learning the joint distribution of the entire video via Flow Matching. Let $\lambda \in [0, 1]$ denote the continuous flow time, and let $\mathcal{X}^\lambda = \{X_1^\lambda, \ldots, X_T^\lambda\}$ be the video state at flow time λ. Starting from Gaussian noise $\mathcal{X}^0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we learn a neural vector field u_θ that transports $\mathcal{X}^0$ to $\mathcal{X}^1$. The generative process follows a probability-flow ODE:

$$\frac{d\mathcal{X}^\lambda}{d\lambda} = u_\theta\big(\mathcal{X}^\lambda, \lambda \mid \mathcal{I}, \mathcal{L}, \mathcal{C}, \mathcal{M}\big), \quad \mathcal{X}^1 = \mathcal{X}^0 + \int_0^1 u_\theta\big(\mathcal{X}^\lambda, \lambda \mid \mathcal{I}, \mathcal{L}, \mathcal{C}, \mathcal{M}\big)\, d\lambda, \quad (1)$$

where $\mathcal{M}$ denotes spatial memory. Compared to the standard TI2V setting, we introduce richer conditional control, most notably through memory retrieval and camera trajectories. We first review the two dominant spatial memory paradigms (explicit and implicit), highlighting their practical limitations. Building on these insights, we introduce *Mosaic Memory*, a novel design that transcends these paradigms while combining their complementary strengths. Furthermore, we develop an improved camera-control module tailored to modern DiT architectures.

2.1 Preliminaries on Spatial Memory: Explicit vs. Implicit

Explicit spatial memory makes the world state *explicit* by *lifting* information from 2D observations into an external 3D geometric cache [4, 7, 28], where the basic storage unit is a set of 3D primitives (e.g., points, voxels, or 3D Gaussian splats shown in Fig. 1) rather than images. Upon revisits, memory retrieval is *optics-based*: the cached 3D structure is *projected* or *rendered* into the queried viewpoint to produce view-aligned conditioning signals, which are typically injected into the generator through mechanisms such as ControlNet-style branches [37] or channel concatenation [28]. The downstream video generator therefore behaves largely as *video inpainting*, filling uncertain or unseen regions while being anchored by projected, geometry-consistent evidence. While this

Figure 2: Shows the complete MosaicMem pipeline: patch lifting into 3D, mosaic composition in target view, and architecture with PRoPE conditioning and warping mechanisms.

Limitations

- Relies on off-the-shelf 3D estimators (Depth Anything V3, VIPE) whose accuracy affects memory quality
- Warped RoPE can cause object repetition at image boundaries in extreme scenarios like very slow camera motion
- Limited to patch-level granularity which may miss fine-grained spatial details within patches
- Memory alignment through warping introduces computational overhead compared to direct frame-based approaches